

PROJECT TITLE

Brita's Water Monitoring Buddy

STUDENT/TEAM INFORMATION

Team Name if any: Team 3	The Water_Boys
Team member 1 (Team Lead)	Abayon, Raymond; rabayon9034@sdsu.edu  A person in a pink Power Ranger suit with a black and white helmet, striking a pose with one hand on their hip and the other pointing upwards.
Team member 2	Wu, Andrew - awu4648@sdsu.edu  A person in a yellow and white Power Ranger suit with a yellow helmet, holding a silver weapon and striking a dynamic pose.

ABSTRACT (15 points)

(Summarize your project (motivation, goals, system design and results). Max 300 words).

The motivation of this Water Monitor project is to give users real-time data for each refill on the Brita pitcher instead of relying on green, yellow, and red light indicators. By having esp32 send pH and turbidity values to the cloud, we are able to send real-time information on water quality and remaining refills left on the Brita pitcher by using our telegram water bot.

INTRODUCTION (15 pts)

Motivation/Background (3 pts)

(Describe the problem you want to solve and why it is important. Max 300 words).

The motivation of this project was quite simple. When I filled myself a glass of water from a Brita pitcher, I drank about half the glass and noticed the Brita pitcher flashing a red light. I started second guessing myself on abomination I just injected into my body. I lost appetite and tossed the water in the sink. That's when this idea came to life. There needs to be a better way than using 3 different light indicators to determine the water quality I am consuming. This is how we came up with using real-time messaging for customers to determine the water quality and remaining refills left on the pitcher.

Project Goals (6 pts)

(Describe the project general goals. Max 200 words).

- Connect esp32 independently to the internet
- Light sensors to indicate water quality on bread board (we are not removing the option of using light indicators but adding more accurate results for users/customers)
- Extract and convert pH and turbidity (analog to digital)
 - Find a usable library for voltage conversion
- Find a messaging API (Telegram)
 - Tried using WhatsApp but their bot worked about 20 percent of the time
- Send pH values and turbidity values to the cloud (AWS)
- Server.py in the cloud
 - Global Array of remaining refills left (46 according to Brita for brand new filter)
 - Sends a message via telegram bot on quality of the water based on refills remaining
 - Web pages for both current values of the current water value and refills
- Send reading to the front end for real-time graphing

Assumptions (3 pts)

(Describe the assumptions (if any) you are making to solve the problem. Max 180 words).

- Each reading is assumed to be one refill of a Brita Pitcher from empty to full
- We are assuming user has an account with Telegram, a Brita pitcher, stable internet, and access to a phone
- User is interested in having the best quality water possible

SYSTEM ARCHITECTURE (20 pts)

(Describe the final architecture you have implemented listing sensors, communication protocols (Wi-Fi, BLE, ...), cloud services and user interfaces. Include a block diagram of the system. Max 300 words).

Write your answer here

Example block diagram:

```
graph TD
    ESP8266[ESP8266] --> WiFi[Wi-Fi]
    ESP8266 --> LEDs[3 LEDs]
    ESP8266 --> pH[pH Probe]
    ESP8266 --> Turbidity[Turbidity Sensor]
    ESP8266 --> AWS[AWS]
    AWS --> Flask[Python Flask]
    AWS --> MobileApp[Mobile App]
```

FINAL LIST OF HARDWARE COMPONENTS (5 pts)

(Write the final list and quantity of the components you have included in your system)

Component/part	Quantity
3 LEDs	3
PH Probe Sensor (SEN0161-V2)	1
Turbidity Sensor (SEN0189	1

LilyGo TTGO ESP32	1
Resistors (330Ω)	3
Wires	12

PROJECT IMPLEMENTATION (30 PTS)

Tasks/Milestones Completed (15 pts)

(Describe the main tasks that you have completed in this project. Max 250 words).

Task Completed	Team Member
- Research on devices needed - Parts have been ordered and have arrived - Circuit Diagram completed - Hardware assembled - Researching acceptable pH and turbidity - Code the esp32 for extraction of data - Began cloud side implementation - Completed Cloud server - Telegram Bot	- Andrew - Andrew/Raymond - Raymond - Raymond - Andrew - Raymond - Andrew - Andrew/Raymond - Andrew

Challenges/Roadblocks (5 pts)

(Describe the challenges that you have faced and how you solved them if that is the case. Max 300 words).

- Difficult to get an accurate reading on turbidity could be due to light being refracted from environment - We did multiple readings and concluded that working in an average lit room resulted in the best results - Finding relationships between analog readings and voltages since a library for our components was not available - Had to do heavy research on Arduino forums for these the pH and turbidity sensors

Tasks Not Completed (5 pts)

(Describe the tasks that you originally planned to complete but were not completed. If all tasks were completed, state so. Max 250 words).

Task	Reason
All Task were completed	

WEAK POINTS / FUTURE WORK (15 pts)

(Mention at least two points of your project that have room for improvement. These points can be additions to the existing project setup (components) or improvement of the current implementation. Max 200 words).

- I believe the weak points of this project would have to be the hardware
 - The turbidity would have to work in an ideal environment, where the room cannot be too dark or too bright.
- Cost
 - Total cost for all the components for my partner and I was about 80 dollars
 - Average Brita user probably does not want to pay more than 80 dollars for real-time water notifications
- User friendly
 - In the future, I would love to input RFID into the pitcher to connect to internet and water messaging system

SOURCE CODE (25 pts)

Please include a link to the source code of your project. A link to a repository (like [GitHub](#)) is preferred.

<https://github.com/Leimondu/IoTProjects/tree/main/FinalProject>